Predicting Credit-Card Default: Eleven Classifiers, One Protocol

Threshold choice matters more than the model; chosen on training CV, the test set only reports.

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Contribution (draft, to confirm): RUAN 29% — KNN benchmark, logistic family, clustering, nested-CV pass · LU 29% — tree, forest, boosting, SHAP/PDP/monotone · LI 42% — SVM/MLP/discriminants, imbalance, leakage, protocol and freeze, cost analysis, deck. Per-section table in the report.

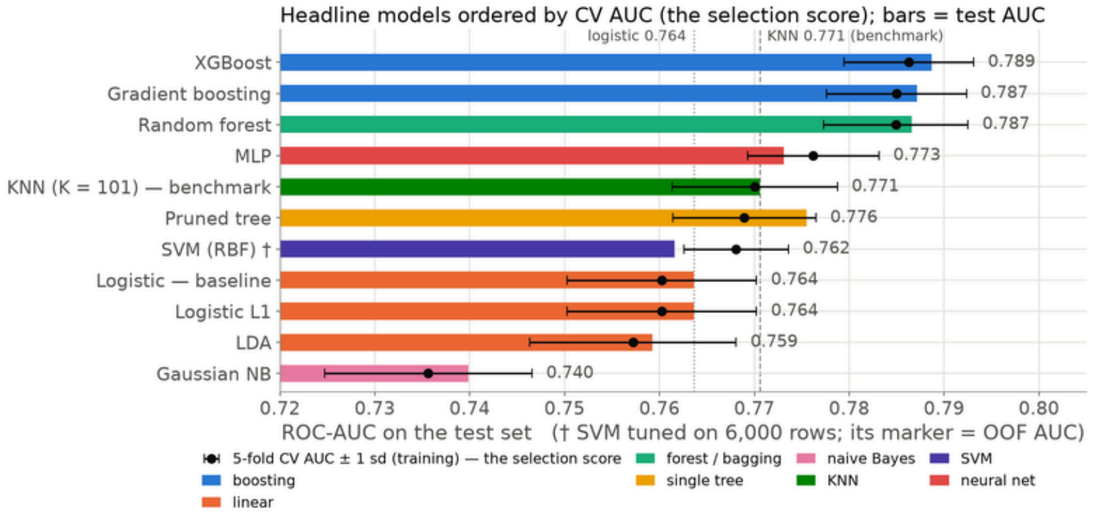


Figure: test AUC (bars) with 5-fold CV standard deviation, models ordered by training CV AUC; results/aggregate_summary.json.
Data: UCI Default of Credit Card Clients (Yeh & Lien 2009), doi:10.24432/C55S3H, CC BY 4.0; 29,965 clients after de-duplication, seed 2607526.

Data & Problem: 29,965 Clients, 22.1% Default Next Month

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Accuracy is not the yardstick — “never default” already scores 0.779.

29,965

CLIENTS

35 duplicates dropped from 30,000
no missing cells; Taiwan 2005

22.1%

DEFAULT RATE

6,630 of 29,965 clients
majority-class accuracy 0.779

29

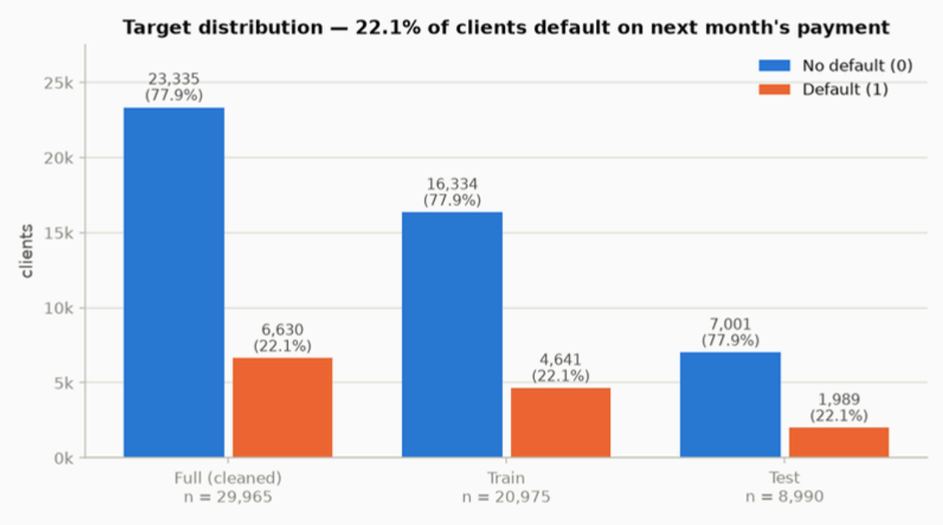
PREDICTORS

23 raw + 6 engineered: utilisation,
pay ratio, delay counts, bill trend

70/30

ONE SPLIT

20,975 train / 8,990 test
seed 2607526, saved to disk



Yardsticks: ROC-AUC and average precision (random = 0.221), never accuracy alone.
Errors are priced: a missed defaulter costs 5 false alarms — one point of a 1:1–20:1 range swept later.

results/eda_summary.json, eda_data_quality.csv, data/processed/split_meta.json; UCI Default of Credit Card Clients (Yeh & Lien 2009), doi:10.24432/C55S3H, CC BY 4.0.

EDA: Repayment Behaviour Is the Signal, Demographics Are Noise

Two months late → 67.6% default, revolving → 12.9%: a step no straight line draws.



67.6% default when two months late in September — vs 33.9% one month late and 12.9% for revolving clients

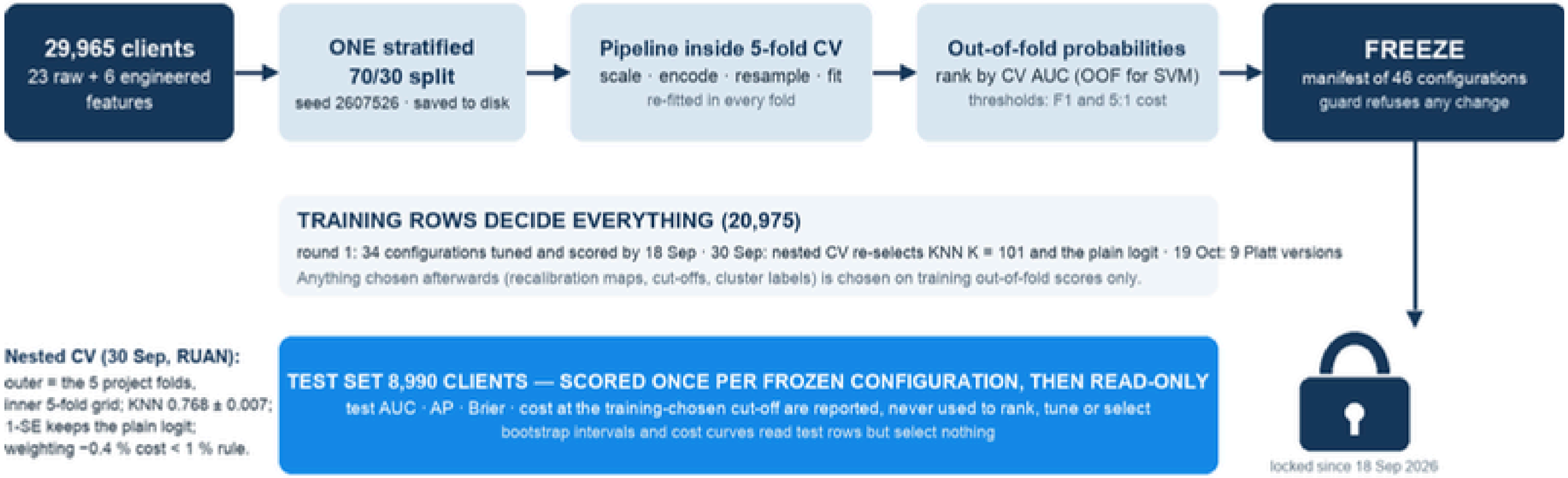
Credit limit protects: 32.1% default in the lowest quintile, 13.7% in the highest · months delayed alone: AUC 0.724 · utilisation U-shaped: 24% → 14% → 28%

→ Behaviour beats demographics; the step and the U are shapes trees can draw and a straight line cannot.

Training partition only (n = 20,975; the test set was not inspected); results/eda_summary.json (default_rate_by, univariate_auc); figures/slides/s_eda_gradients.png.

Protocol: Chosen on Training CV, Frozen, Scored Once on Test

Every choice on 20,975 training rows; 46 frozen configurations; the test set only reports.



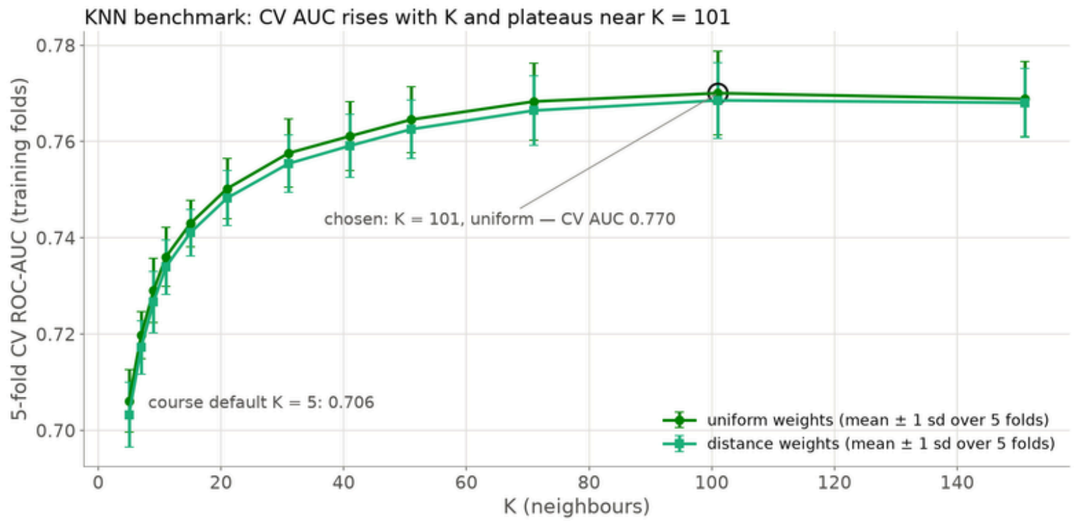
Rule: models ranked by training CV AUC, cut-offs from training out-of-fold scores; a changed configuration is refused before any test label is read.

results/frozen_manifest.json (46 entries: 34 by 18 Sep, 2 adopted from the 30 Sep pass, 9 Platt on 19 Oct, 1 leakage control), nested_cv_summary_repro.json, knn_v2.json; code/freeze.py --check passes.

Benchmark and Linear Family: KNN 0.770, Logistic 0.760

KNN with K = 101 is the bar; plain, lasso and ridge logistic all sit one point below it.

- KNN K = 101, uniform (benchmark): CV 0.770 (sd 0.009) · test 0.771 · AP 0.542
- KNN K = 5: CV 0.706 · test 0.709 — one noisy neighbour dominates each vote
- Logistic plain / L1 / L2: CV 0.760 · test 0.764 · AP 0.523 — shrinkage buys nothing
- Nested CV (30 Sep): KNN 0.768 ± 0.007, K = 101 re-selected; 1-SE rule keeps the plain logit
- Class weighting: −0.4% cross-fitted 5:1 cost, below the 1% rule → not adopted



Rule: every later model must beat KNN 0.770 on training CV — or explain why it does not.

results/knn.json (cv_table), knn_k5.json, logistic.json, logistic_family_summary.csv, nested_cv_summary_repro.json; CV = 5-fold on 20,975 training rows; test n = 8,990; figure: figures/slides/s_knn_k.png.

Tree Family: Pruning, Averaging and Boosting Climb to CV 0.786

One pruned tree ties KNN; 500 averaged or 300 boosted depth-3 trees add 0.016–0.017 CV AUC.

ONE TREE: 2,791 → 17 LEAVES

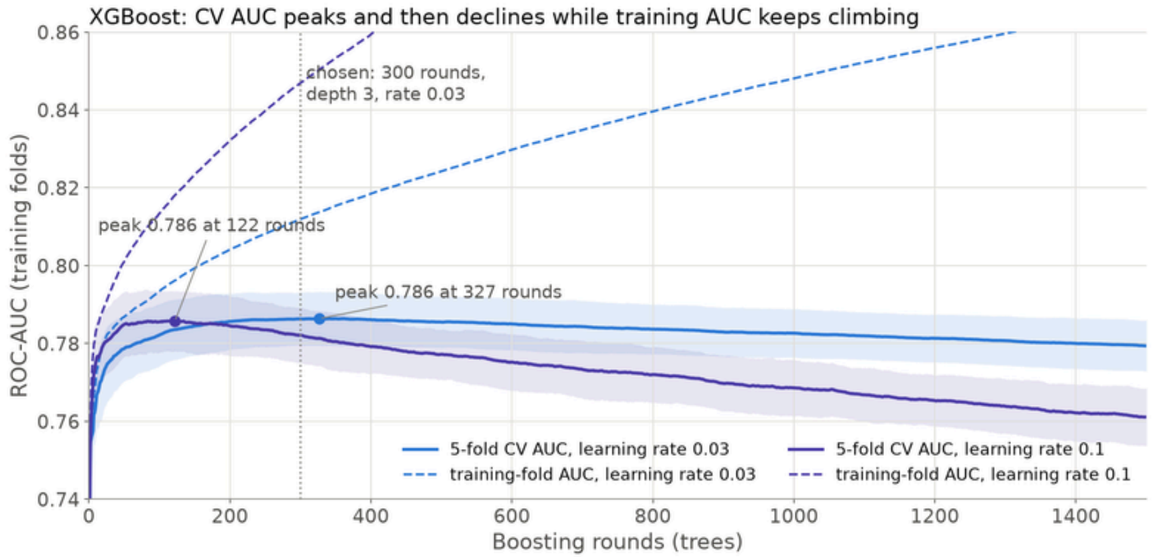
Grown to purity: CV AUC 0.615 (memorises) → pruned on training CV: CV 0.769, test 0.776; root split MAX_DELAY > 1.5

RANDOM FOREST

500 trees, ≥ 20 clients per leaf: CV 0.785, test 0.787 — averaging removes the variance a single tree memorises

XGBOOST — CV WINNER

depth 3, rate 0.03, 300 rounds: CV 0.786, test 0.789, AP 0.578 — fits in 0.5 s against 19.7 s for the forest

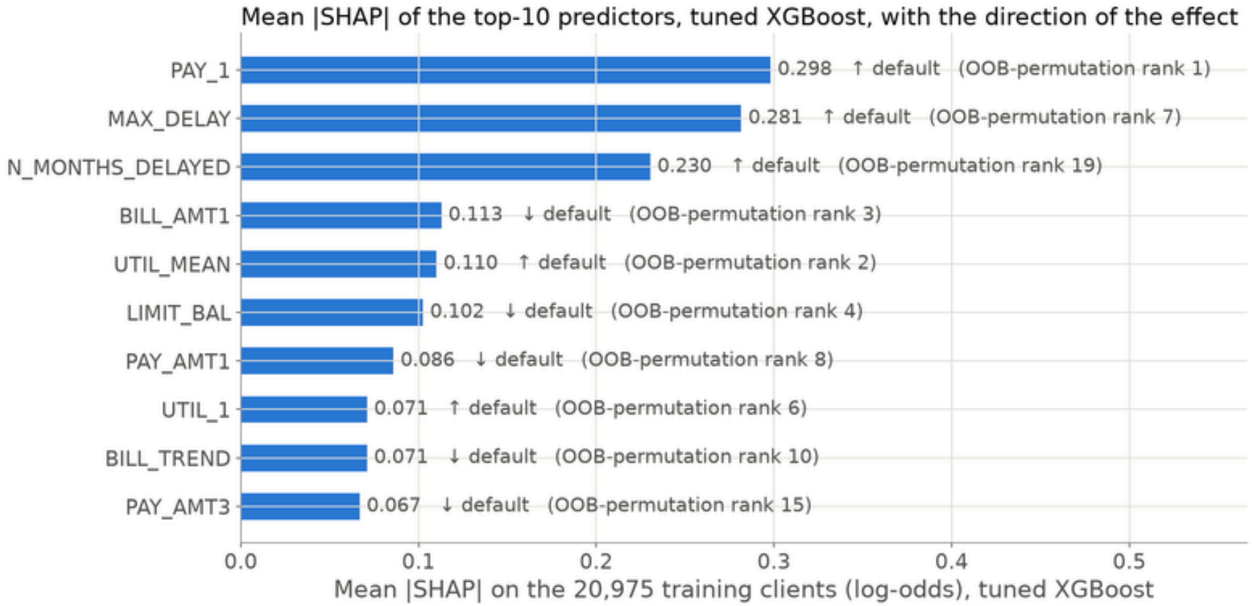


XGBoost − forest +0.002 [−0.001, +0.005] on the paired bootstrap: a tie — XGBoost kept as CV winner, forest as fallback.

results/tree_full.json, tree_pruned.json, tree_rules.txt, random_forest.json, xgboost.json, xgb_learning_curve.csv, auc_bootstrap_ci.csv (B = 500); CV = 5-fold on 20,975 training rows; test n = 8,990; figure: figures/slides/s_xgb_learning_curve.png.

Interpretation: Repayment Delay Drives the Score

Three delay features carry the model; monotone constraints remove 24 wrong-direction steps for −0.002 AUC.



- TreeSHAP on the 20,975 training rows: PAY_1 0.298 · MAX_DELAY 0.281 · N_MONTHS_DELAYED 0.230; bills, utilisation and limit at a third of that
- Drop-column refit of the forest agrees: removing PAY_1 costs the most out-of-bag AUC, 0.785 → 0.779
- Partial dependence keeps the utilisation U: minimum 0.204 at utilisation 0.08
- Monotone XGBoost (11 features up, 2 down): wrong-direction steps 24 → 0; test AUC 0.789 → 0.787
- Beyond the syllabus — XGBoost, TreeSHAP, PDP/ICE, monotone constraints, Platt, paired bootstrap — used to explain, never to select

results/shap_importance.csv (TreeSHAP on training rows only; test set not read), rf_refit_drop_column.csv (OOB), pdp_summary.json, monotone_summary.json; Lundberg & Lee (2017); figure: figures/slides/s_importance.png.

SVM, MLP, Discriminants: Only the Network Clears the Benchmark

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MLP CV 0.776 beats KNN 0.770; RBF SVM 0.761, LDA 0.757, QDA 0.751 and NB 0.736 fall below.

	MODEL	MECHANISM	SELECTION AUC · TEST AUC · AP
01	MLP 32-64-32-1, EARLY STOP	logistic plus learned kinks; 4,225 weights	**CV 0.776 · test 0.773 · AP 0.551 — above KNN** no-stopping twin (74,497 weights): held-out AUC 0.760 at epoch 1 → 0.700 at 200; CV 0.692
02	RBF SVM C = 1, γ = 0.01	margin loss ranks the tail poorly; tuned on 6,000 rows	**OOF 0.761 · test 0.762 · AP 0.494 — below KNN** subsample CV 0.768 not comparable; Platt probabilities; 47 s refit
03	LDA / QDA	Gaussian class densities on skewed amounts	**CV 0.757 / 0.751 · test 0.759 / 0.755 — below KNN** AP 0.525 / 0.516; LDA tracks the logit direction (cosine 0.93); QDA shrinks covariances to the grid edge
04	GAUSSIAN NAIVE BAYES	independence fails: six bill columns correlate at 0.80–0.95	**CV 0.736 · test 0.740 · AP 0.504 — lowest; Brier 0.203**

results/mlp.json, mlp_overfit.json, svm_rbf.json, lda.json, qda.json, naive_bayes.json, eda_summary.json; selection = 5-fold CV AUC on training rows (OOF AUC for the subsample-tuned SVM); test n = 8,990.

Imbalance: Re-sampling Moves the Threshold, Not the Ranking

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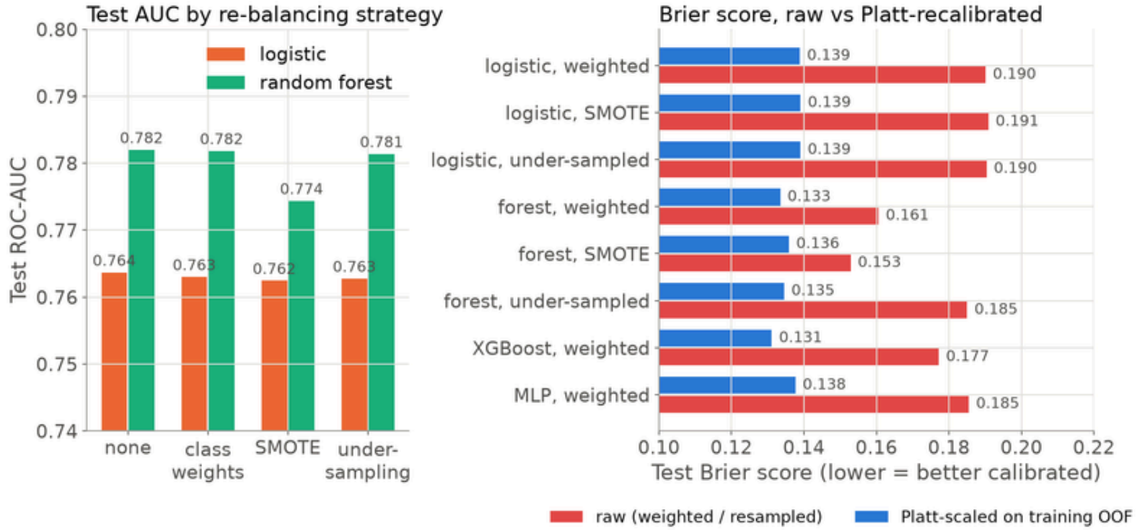
Four strategies move logistic test AUC by about 0.001; Platt on training OOF restores calibration.

WHAT WE COMPARED

none · class weights · SMOTE · random under-sampling, inside the folds only for logistic regression and random forest (eight runs, frozen 16–18 Sep)

WHAT HAPPENED

- Logistic test AUC 0.764 → 0.762 across all four; forest with SMOTE 0.782 → 0.774
- Balanced logit at 0.5 = plain logit at 0.22 on 99.4% of test clients
- Weighting breaks calibration: mean p 0.44 vs prevalence 0.22; Brier 0.138 → 0.190



Our fix: keep the natural prior, choose the cut-off on training OOF scores, Platt-recalibrate anything weighted.

Platt on training OOF returns weighted XGBoost's Brier 0.177 → 0.131 and the 5:1 cut-offs to 0.14–0.19; all nine versions scored once on 19 Oct and frozen.

results/imbalance_comparison.csv, imbalance_threshold_equivalence.csv, recalibration.csv (nine Platt versions frozen 19 Oct); samplers inside the folds (imblearn Pipeline); Chawla et al. (2002); Platt (1999); test n = 8,990.

Leakage Control: SMOTE Before the Split Fakes 0.914 AUC

Honest forest 0.774; the same forest “tested” on 73% synthetic defaulters scores 0.914.

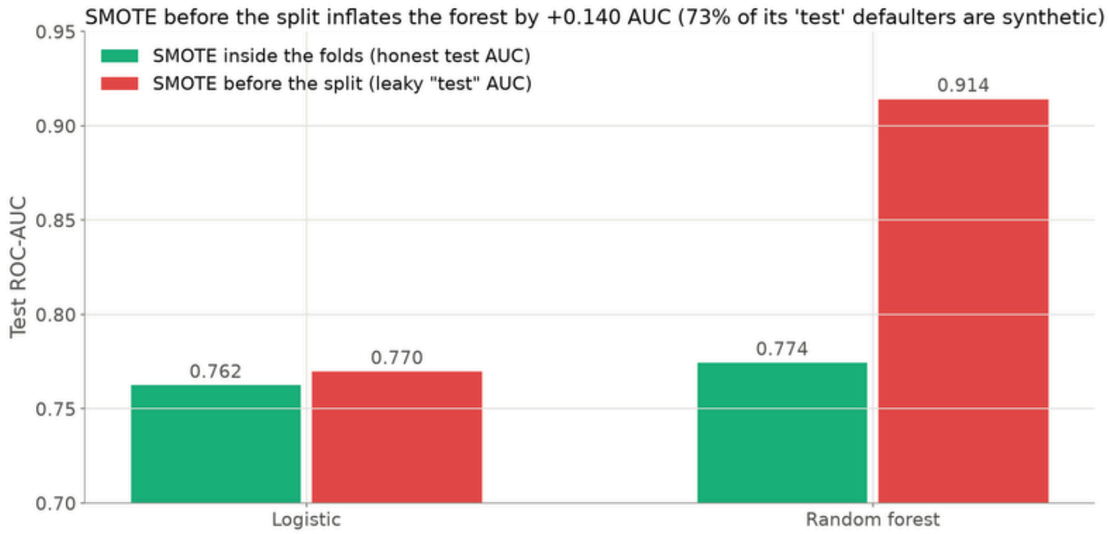
WHY IT LEAKS

SMOTE first, split second: synthetic defaulters land on both sides

73% of the leaky test defaulters are copies of training defaulters

The forest recognises its own neighbours; a straight line cannot memorise

SAME LEARNERS, TWO ORDERS OF OPERATIONS

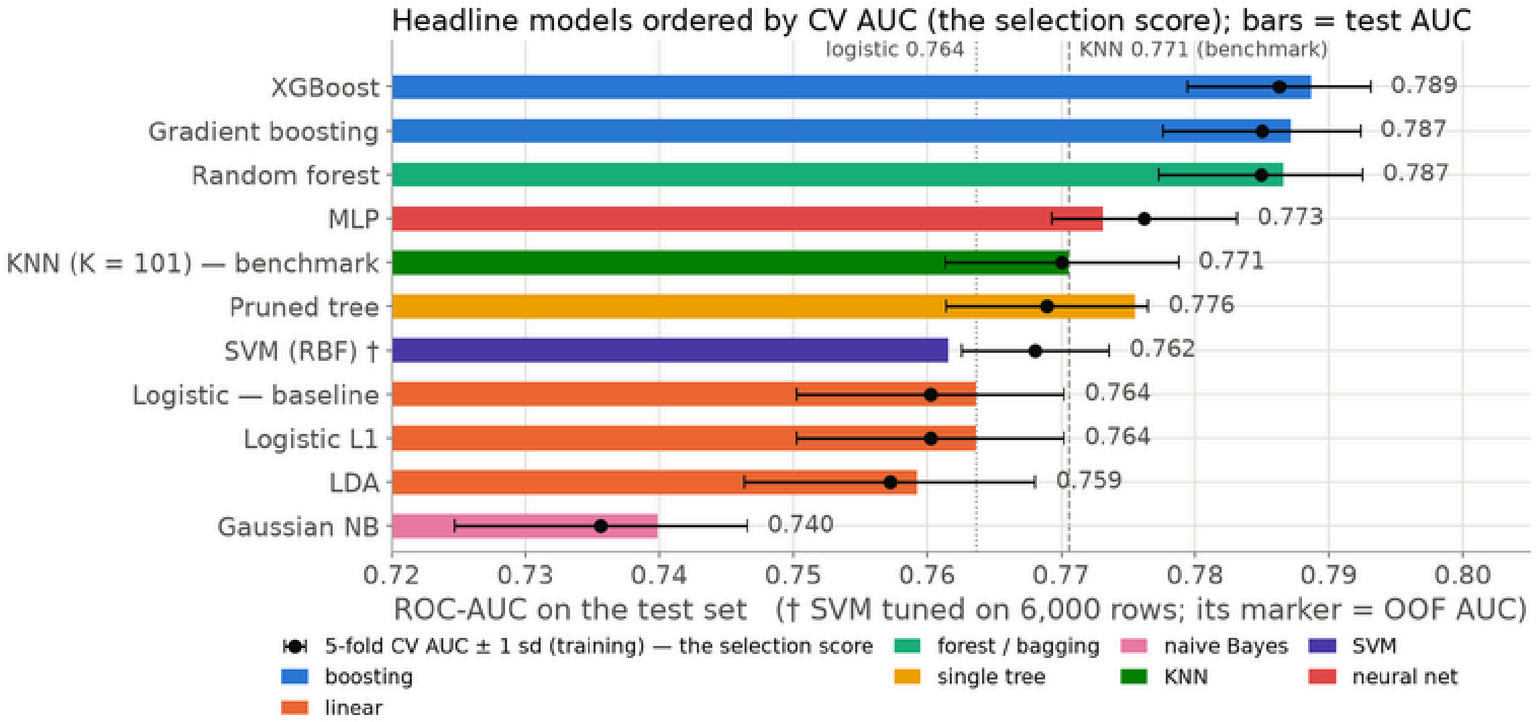


Forest: honest 0.774 → leaky 0.914 (+0.140); logistic 0.762 → 0.770 (+0.007) — leakage flatters the flexible model most.

results/imbalance_leakage_demo.csv — quarantined negative control (slug imb_leaky_smote_demo in frozen_manifest.json); 16,705 synthetic rows; honest runs keep every sampler inside the fold.

Ranking by Training CV: Boosting Leads Logistic by 0.025

Above KNN: XGBoost, GBM, forest, MLP; the pruned tree ties it; SVM, logistic, LDA, NB fall below.



1 · Above the benchmark

XGBoost CV 0.786 · test 0.789 · AP 0.578

2 · Tied with KNN

pruned tree CV 0.769 vs 0.770

3 · Paired bootstrap

vs logistic +0.025 [+0.018, +0.031]
vs KNN +0.018 [+0.012, +0.024]

4 · Top three are one group

XGB − forest +0.002 [−0.001, +0.005]

Order = training CV AUC (the selection score; OOF AUC for the subsample-tuned SVM) · bars = test AUC on 8,990 clients, markers = CV ± 1 sd · results/aggregate_summary.json, model_comparison_main.md, auc_bootstrap_ci.csv (B = 500).

Operating Point: the Cut-off Saves 1,937, the Model 145

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At 5:1, re-thresholding logistic saves 1,937 [1,604, 2,218]; switching to XGBoost saves 145 [−18, 299].

1,937

CUT-OFF SAVING

logistic cut-off 0.5 → 0.15 at 5:1
95% CI [1,604, 2,218]

145

MODEL SAVING

logistic → XGBoost, own cut-offs
95% CI [−18, 299] includes 0

0.15

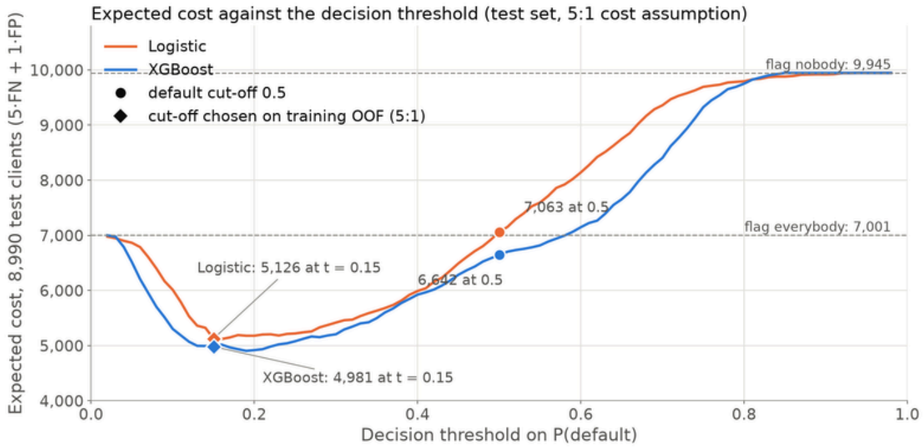
5:1 CUT-OFF

chosen on training OOF scores
closed form 1/(1+5) = 0.17

20:1

DEGENERATE END

optimal cut-off 0.01 flags 99.9%
the trivial rule, not a model result



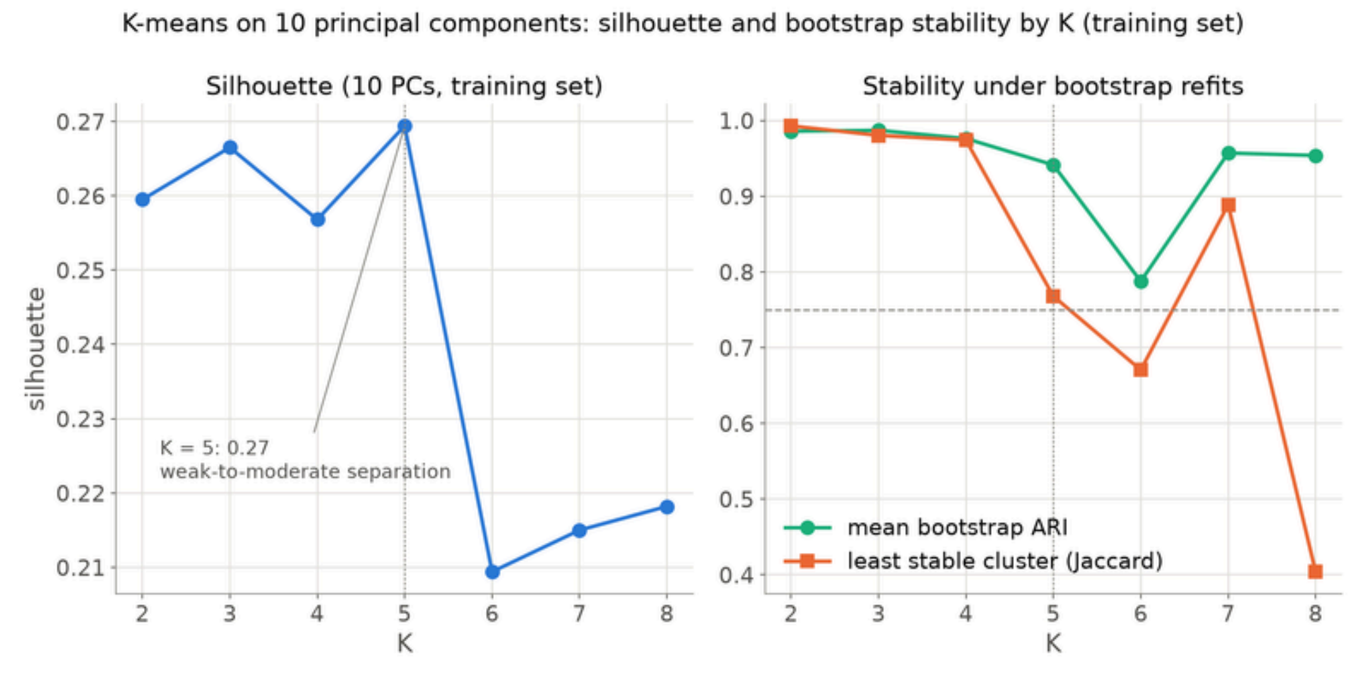
5:1 is one point of a 1:1–20:1 sweep: the threshold saving grows with the ratio (684 at 3:1, 7,121 at 10:1), the model saving stays in the low hundreds (87–420). Fix the cut-off from the cost ratio first; the model comes second.

results/threshold_vs_model_saving.csv (cut-offs on training OOF; paired bootstrap B = 1,000 on the 8,990 test rows, reporting only), cost_sensitivity.csv; logistic cost 7,063 at 0.5, 5,126 at 0.15; XGBoost 4,981 at 0.15; figure: figures/slides/s_cost_vs_threshold.png.

Business: a Map for Policy, a Cut-off for Decisions

K-means K = 5 on 10 PCs is stable but adds no AUC: it describes clients, it does not score them.

CLUSTER STABILITY (TRAINING ROWS ONLY)



WHAT WE FOUND

- PCA: 10 PCs = 81.8% of variance; K = 5, silhouette 0.269 (weak-to-moderate)
- Stable: bootstrap ARI 0.957 [0.745, 0.990], Jaccard 0.79–0.97; K-sweep with fewer draws (figure): 0.941 / 0.768
- Cluster label as extra predictor: ΔAUC and ΔAP below 0.001, intervals include 0 → not adopted
- Deploy: XGBoost or forest ranking; cut-off from the cost ratio; Platt if weighted
- Explain: logistic or the 17-rule tree; monotone XGBoost where regulation demands

A map for policy, not a better model: rank with XGBoost, cut off by the cost ratio, recalibrate anything weighted.

results/unsupervised_summary_repro.json, kmeans_profiles_repro.csv, kmeans_selection_repro.csv, cluster_incremental_folds_repro.csv (RUAN's 30 Sep analysis reproduced 21 Oct on training rows; the test set was not consulted).

Limitations and Next Steps: One Split, One Assumed Cost Ratio

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The ranking is honest but narrow: one split, one seed, a 2005 Taiwan sample and an assumed 5:1.

01

WHAT WE FOUND

XGBoost CV 0.786, test 0.789: top three tied

The cut-off saves 1,937; the model 145

Re-balancing shifts thresholds, not ranks

Clusters describe clients, they do not score them

02

LIMITATIONS

One 70/30 split, one seed: gaps below 0.005 sit inside bootstrap noise

5:1 assumed; bill-and-margin anchors imply 9.6–40, hence the sweep

Taiwan 2005, one-month horizon; PAY_1 observed one month before the label

03

NEXT STEPS

Repeated splits; nested CV for every family

Loss-given-default pricing of the errors

Drift monitoring with periodic recalibration

Per-client uncertainty intervals

Appendix · Nested CV, Platt, Cost Sweep, Settings and References

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NESTED CV (RUAN, 30 Sep; reproduced 21 Oct on training rows only) — outer = the five project folds, inner 5-fold grid
KNN K = 101 uniform: nested AUC 0.768 (sd 0.007) → frozen choice confirmed · Logistic plain 0.760 (SE 0.005), 5:1 OOF cut-off 0.17, cross-fitted cost 12,195 → kept by the 1-SE rule · ridge 0.760 / lasso 0.760 within 1 SE · class-weighted 0.761, mean p 0.44 (mis-calibrated) · weighted + Platt 0.761, cost 12,247: −0.4% < 1% rule → not adopted

PLATT RECALIBRATION (19 Oct, nine weighted or resampled models) — maps fitted on training out-of-fold scores, thresholds cross-fitted; Brier returns to the unweighted twins (weighted XGBoost 0.177 → 0.131); 5:1 cut-offs 0.14–0.19; each calibrated version scored once on the test set and frozen

COST SWEEP (threshold saving vs model saving, paired bootstrap B = 1,000 on the 8,990 test rows) — 1:1: 0 vs 87 [44, 131] · 3:1: 684 [541, 830] vs 127 [47, 214] · 5:1: 1,937 [1,604, 2,218] vs 145 [−18, 299] · 10:1: 7,121 [6,395, 7,791] vs 420 [187, 646] · 20:1: 20,335 [18,909, 21,621] vs 150 [−52, 333], cut-off 0.01 flags 99.9%

TUNED SETTINGS (5-fold CV AUC on 20,975 training rows, seed 2607526) — XGBoost depth 3, learning rate 0.03, 300 rounds · Random forest 500 trees, ≥ 20 clients per leaf, sqrt features · Pruned tree 17 leaves (cost-complexity path) · KNN K = 101 uniform · Logistic plain; L1/L2 at their CV optimum keep all 32 columns · RBF SVM C = 1, γ = 0.01, balanced weights (6,000-row tuning subsample) · MLP 32-64-32-1 with early stopping

REFERENCES — Yeh, I.-C. & Lien, C.-H. (2009). Expert Systems with Applications 36(2) · James, Witten, Hastie & Tibshirani, An Introduction to Statistical Learning · Breiman, L. (2001). Random forests. Machine Learning 45 · Chen, T. & Guestrin, C. (2016). XGBoost. KDD · Chawla, N. V. et al. (2002). SMOTE. JAIR 16 · Lundberg, S. & Lee, S.-I. (2017). A unified approach to interpreting model predictions. NeurIPS · Platt, J. (1999). Probabilistic outputs for support vector machines · Hofmann, H. (1994). Statlog German Credit, UCI · Pedregosa, F. et al. (2011). Scikit-learn. JMLR 12 · Code, data and results: github.com/jingwen013-design/mh6805-credit-default